

CLOCK-BASED INTERVIEW QUESTIONS

RTL Design • VLSI • Design Verification • Clocking • CDC • Timing

Purpose

A practical interview-preparation resource covering clock fundamentals, frequency and duty-cycle calculations, clock dividers, clock gating, clock muxes, timing, CDC, synchronizers, asynchronous FIFO concepts, PLLs and CTS.

1. Clock Fundamentals

Topic	Key point
Frequency	Number of clock cycles per second.
Time period	$T = 1 / f$
Duty cycle	HIGH time / total period $\times 100\%$
posedge	$0 \rightarrow 1$ transition
negedge	$1 \rightarrow 0$ transition
Clock skew	Difference in clock arrival time between endpoints.
Clock jitter	Variation of an edge from its ideal timing position.
Clock latency	Time taken for the clock to reach a destination.

2. Calculation Questions

Formula: $T = 1/f$

Examples: 100 MHz $\rightarrow$ 10 ns. 20 ns $\rightarrow$ 50 MHz.

Duty-cycle example: 20 ns period with 15 ns HIGH time $\rightarrow$ 75% duty cycle.

3. Clock Divider Questions

Must-practice RTL problems: divide-by-2, divide-by-4, divide-by-8, divide-by-10, divide-by-3, divide-by-5, divide-by-7, divide-by-16, programmable divide-by-N, and odd dividers with controlled duty cycle.

Basic F/2 divider:

```
reg clk_div2;
always @(posedge clk or posedge rst) begin
    if (rst) clk_div2 <= 1'b0;
    else clk_div2 <= ~clk_div2;
end
```

4. Clock Enable vs Generated Clock

A clock enable lets logic remain in the original clock domain while updating only when enabled. This is often preferable to creating many derived clocks in RTL because it simplifies timing and CDC analysis.

Example 1: update a register every 4 cycles using an enable.

Example 2: generate a one-cycle pulse every 10 cycles with a counter.

5. Clock Gating

Clock gating is used to reduce dynamic power by preventing unnecessary clock activity. A naive expression such as **assign gated_clk = clk & enable;** can create glitches if enable changes while the clock is active.

Interview focus: Why can a combinational AND gate create a bad clock? What makes a clock-gating structure glitch-free? Why are integrated clock-gating cells commonly used in ASIC flows?

6. Clock MUX

Selecting between two clocks with a normal combinational MUX can generate glitches or shortened pulses when the select signal changes. Interviewers often ask for a glitch-free clock mux concept using mutually controlled clock gates.

7. Timing: Setup, Hold, Skew and Jitter

Concept	Interview definition
Setup time	Data must be stable for a minimum time before the capture edge.
Hold time	Data must remain stable for a minimum time after the capture edge.
Positive skew	Capture clock arrives later than launch clock; generally helps setup but can hurt hold.
Negative skew	Capture clock arrives earlier; generally hurts setup but can help hold.
Jitter	Temporal movement/variation of clock edges.

8. Clock Domain Crossing (CDC)

CDC is required when a signal crosses between clock domains without a guaranteed safe timing relationship.

2-Flip-Flop Synchronizer

```
async_data → FF1 → FF2 → synchronized_data
```

The first flip-flop may become metastable if the asynchronous signal changes near the destination clock edge. The second flip-flop provides additional resolution time before the signal reaches normal logic.

Important: A 2-FF synchronizer is appropriate for a single-bit level signal. It is not a general solution for transferring a multi-bit data bus.

9. Multi-Bit CDC

Common techniques include handshake synchronizers, asynchronous FIFOs, Gray-coded counters and protocol-based data transfer.

Example 1: Synchronize a single interrupt/status bit with a 2-FF synchronizer.

Example 2: Transfer a stream of 32-bit data between unrelated clocks using an asynchronous FIFO.

10. Asynchronous FIFO Clock Questions

Key interview topics: independent write/read clocks, binary pointers, Gray-code pointers, 2-FF pointer synchronizers, full detection, empty detection and safe CDC.

Why Gray code? Binary counters can change multiple bits in one transition, while Gray code changes only one bit at a time, making synchronized pointer transitions safer to interpret.

11. PLL and Clock Generation

A PLL can be used for clock multiplication, division, phase alignment and clock generation.

Examples: $50 \text{ MHz} \times 4 = 200 \text{ MHz}$. $100 \text{ MHz} \div 2 \times 5 = 250 \text{ MHz}$.

12. Clock Tree Synthesis (CTS)

CTS builds the physical clock distribution network from clock sources to sequential elements while controlling skew, latency, transition and power. Clock networks require special treatment because clock quality directly affects timing closure.

13. Reset and Clock Questions

Know the difference between synchronous and asynchronous reset. A common advanced question is why asynchronous reset assertion may be acceptable while asynchronous deassertion can be dangerous. Reset release is often synchronized to avoid inconsistent recovery.

14. High-Value Interview Coding Problems

1. Generate a 100 MHz clock in a testbench.
2. Design F/2, F/4 and F/8 dividers.
3. Design divide-by-N programmable clock divider.
4. Generate a one-cycle pulse every N cycles.
5. Generate a clock enable every N cycles.
6. Design glitch-free clock gating concept.
7. Design a glitch-free clock mux concept.
8. Design a 2-FF synchronizer.

9. Design a pulse/toggle synchronizer.
10. Design a handshake CDC.
11. Explain and implement an asynchronous FIFO.
12. Analyze setup/hold violations caused by clock skew.

15. Interview Trap Questions

- Why is assign gated_clk = clk & enable unsafe?
- Why is a normal MUX unsafe for clock switching?
- Why can't 32 independent 2-FF synchronizers safely transfer a 32-bit bus?
- Why is a generated clock often less desirable than a clock enable in RTL?
- How does positive clock skew affect setup and hold?
- Why is asynchronous reset deassertion a timing concern?
- What is the difference between clock skew and jitter?
- Why does an asynchronous FIFO use Gray-coded pointers?

16. Recommended Learning Order

Clock basics → Frequency/period → Duty cycle → Clock dividers → Counters → Clock enable → Clock gating → Glitch-free clock mux → Setup/Hold → Skew/Jitter → CDC → 2-FF synchronizer → Handshake CDC → Async FIFO → PLL → CTS.

Prepared as a practical RTL/VLSI interview revision resource.

Focus on understanding the timing behavior and writing RTL, not memorizing definitions.